

EDUCATION			
YEAR	DEGREE / EXAM	INSTITUTE	CGPA / MARKS
2025-27	M.Tech.	Indian Institute of Technology, Tirupati	8.83
2017-20	M.Sc.	Indian Institute of Technology, KANPUR	6.2
2014-17	B.Sc.	University of Delhi, New Delhi	76.69%

PROJECTS

Geo-Spatial Semantic Segmentation | [GITHUB](#)

June-2026

Tools & Technologies: Python, PyTorch, Mask2Former, Computer Vision

- Defined** multi-channel GeoTIFF pixel classification as a dense semantic segmentation problem and built a Mask2Former(small)-based pipeline to solve it, achieving 86.7% Mean IoU (excluding background) and 97.4% pixel accuracy across 8,973 test images.
- Built** an end-to-end inference and georeferencing workflow, including annotation-aware visualization to make model predictions interpretable and verifiable against ground geography.
- Awarded** Innovation Recognition, Ministry of Panchayati Raj (MoPR) Geospatial Intelligence All-India Hackathon, for the solution's applicability to real-world governance data

Intelligent LLM Text Classification Platform| [GITHUB](#)

June-2026

Tools & Technologies: Python, PyTorch, GPT-2, Transformer architecture, Tiktoken, NumPy, Pandas, TensorFlow, Hugging Face Hub

- Modeled** text generation as a next-token probability estimation problem and implemented a GPT-style transformer in PyTorch, covering tokenization through autoregressive decoding.
- Formulated** SMS spam filtering as a binary classification problem GPT-2 (91.88% test accuracy) and benchmarked model performance against BERT (90.89% test accuracy) and RoBERTa (92.95% test accuracy) on IMDb sentiment classification.

Enhanced YOLOv9 for Real-Time Object Detection| [GITHUB](#)

Apr-2026

Tools & Technologies: Python, PyTorch, YOLO-v9, Object Detection, Deep Learning

- Analyzed** baseline detection failure modes and modeled a channel-attention mechanism (CBAM) to address them, integrating three attention modules across multi-scale feature maps of the YOLOv9 architecture.
- Trained** and evaluated the modified architecture on PASCAL VOC (20 classes), achieving 79.2% mAP@0.5 and a 0.39% improvement in detection accuracy over the baseline.
- Built** a diagnostic evaluation pipeline (confusion-matrix analysis, precision-recall/F1 curves) to explain and quantify the source of the improvement

Reimagining Wikipedia in AI era | [GITHUB](#)

Apr-2026

Tools & Technologies: React , Express , Ollama , RAG , Vector Embeddings , Prompt Engineering , YouTube Data API

- Designed** a retrieval-augmented generation architecture to ground free-text responses in verifiable sources, addressing unexplainable/unattributed model outputs.
- Modeled** persona-aware prompting and retrieval logic and integrated LLMs with external APIs into a full-stack application for personalized explanations, quizzes, and summaries.

TECHNICAL SKILLS	
Programming	C++ Python
Libraries/Frameworks	Langchain Langgraph Ollama RAG Vector Embeddings Prompt Engineering PyTorch PyTorch Geometric NumPy Scikit-learn Pandas OpenCV MongoDB
Concepts	Agentic AI RAG Diffusion model transformer ViT VLM LLM GAN

ACHIEVEMENTS	
AIR-85 in IIT JAM MATHS, 2017 Gate Score 731 , AIR-871 GATE CSE 2025 AIR-459 in IIT JAM STATS, 2017 AIR-111 in CSIR NET(Mathematics) in June 2022 Awarded the KIA–IITT Scholarship for academic merit at IIT Tirupati 2026. Innovation Recognition Award, Ministry of Panchayati Raj (MoPR) Geospatial Intelligence all India AI/ML Hackathon 2026.	

CERTIFICATIONS	
Nokia Workshop 2026 on AI, 6G, and Edge Cloud Convergence, exploring next-generation AI-driven network technologies and architectures Received a Certificate of Appreciation for coordinating the 2026 M.Tech. DSAI admission process at IIT Tirupati (GANs) Generative Adversarial Networks - DeepLearning.AI (Coursera)	